THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Fractions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\frac{5}{6} + \frac{3}{8} - \frac{1}{4} =$ 1

(a) $\frac{17}{24}$ **(b)** $\frac{23}{24}$ **(c)** $1\frac{1}{6}$ **(d)** $1\frac{11}{24}$

2. $\frac{3}{4}$ of a number minus $\frac{1}{2}$ of it is 15. The number is: 1

(a) 30 **(b)** 70 **(c)** 12 **(d)** 60

3. A drum is $\frac{3}{4}$ full. 20 litres are removed and it becomes $\frac{1}{4}$ full. The capacity (litres) is: 1

(a) 39 **(b)** 40 **(c)** 41 **(d)** 50

4. A rod 6 m long is cut into pieces of $\frac{3}{4}$ m each. The number of pieces is: 1

(a) 8 **(b)** 10 **(c)** 7 **(d)** 9

5. If $\frac{3}{5}$ of a number is 12, then $\frac{1}{4}$ of the number is: 1

(a) 20 **(b)** 12 **(c)** 5 **(d)** 1

6. Which is the smallest of $\frac{5}{7}$, $\frac{7}{10}$, $\frac{3}{4}$, $\frac{8}{11}$? 1

(a) $\frac{3}{4}$ **(b)** $\frac{8}{11}$ **(c)** $\frac{5}{7}$ **(d)** $\frac{7}{10}$

7. A rod $3\frac{1}{2}$ m long has $\frac{3}{4}$ m cut off. The remaining length in metres is: 1

(a) $2\frac{3}{4}$ **(b)** $2\frac{1}{4}$ **(c)** $4\frac{1}{4}$ **(d)** $\frac{1}{4}$

8. To add $\frac{1}{6}$ and $\frac{1}{8}$, the common denominator used is the LCM of 6 and 8. The LCM is: 1

(a) 48 **(b)** 24 **(c)** 30 **(d)** 12

9. **Assertion (A):** $\frac{3}{4}$ and $\frac{6}{8}$ are equivalent fractions. 1

Reason (R): Multiplying the numerator and denominator by the same non-zero number gives an equivalent fraction.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The fraction $\frac{7}{4}$ is an improper fraction. 1
Reason (R): $\frac{7}{4} = 1\frac{3}{4}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. $\frac{3}{4}$ of a number is 18. Find the number and its $\frac{1}{3}$. 2

12. Find the fraction which must be added to $\frac{2}{7}$ to get $\frac{5}{6}$. 2

13. A boy walks $\frac{3}{4}$ km to school and $\frac{2}{5}$ km to a shop, then returns by the same route. Find the total distance. 2

14. $\frac{2}{5}$ of a number is 14. Find the number and its $\frac{3}{7}$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A school has 480 students. $\frac{5}{8}$ of them are girls. $\frac{2}{5}$ of the girls play chess. How many girls play chess and how many boys are there? 3

16. Find the missing numerator ?: $\frac{?}{12} + \frac{1}{4} = \frac{5}{6}$. 3

17. Pipe A fills $\frac{1}{4}$ of a cistern in one hour and pipe B fills $\frac{1}{6}$ of it in one hour. What part of the cistern is filled in one hour when both work together? What part remains empty? 3

OR

A school has 360 students. $\frac{4}{9}$ of them are girls. $\frac{3}{8}$ of the girls play chess. How many girls play chess and how many boys are there? 3

18. Find the missing numerator ?: $\frac{?}{20} + \frac{1}{5} = \frac{3}{4}$. 3

OR

Pipe A fills $\frac{1}{3}$ of a cistern in one hour and pipe B fills $\frac{1}{5}$ of it in one hour. What part of the cistern is filled in one hour when both work together? What part remains empty? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A pizza is cut into 8 equal slices. Kavita eats $\frac{3}{8}$ of the pizza and her brother Rohan eats $\frac{2}{8}$ of it. 5
(i) What fraction of the pizza did they eat together?
(ii) What fraction is left?
(iii) Who ate more?
(iv) How many slices are left, and what is the fraction left written with denominator 16?

20. A trader sold $\frac{1}{4}$ of his cloth on day 1 and $\frac{2}{3}$ of the remaining on day 2. If 12 metres are left, find the total length of cloth. 5

OR

Add $2\frac{1}{4}$, $1\frac{1}{2}$ and $1\frac{1}{8}$ and subtract $\frac{1}{2}$ from the sum. 5

END OF QUESTION PAPER — **Total: 40 Marks**